

Active Directory Home Lab

Shared Folder Permissions & Mapped Network Drives

Platform: Oracle VirtualBox | OS: Windows Server 2022 + Windows 11

Overview

This section covers how to tie user groups to shared folders and control who can access what data on the network. It also covers two ways to give users access to their shared folders – manually mapping a network drive and automating the process through Group Policy, so it happens on login.

The scenario used throughout: Jake Anthony just started working in HR and needs access to the HR shared folder on the server.

Real World Context

File Servers vs Domain Controllers

In a production environment, shared folders never live on the Domain Controller. They live on a separate dedicated server called a File Server. If the Domain Controller crashes and the shared data is on the same machine, everything goes down together. Keeping them separate means one failure does not take out the other.

In this lab, the shared folder is on the Domain Controller since it is a virtual environment with only one server. This is fine for learning, but not how it would be done in a real company.

Important: In a real environment, always keep shared data on a separate File Server – never on the Domain Controller.

Failover Domain Controllers

Companies always have at least one backup Domain Controller. If the only DC goes down, nobody can log in. The backup is usually hosted in the cloud or a separate location and is called a Failover DC. A common setup is one DC in the office and one in the cloud.

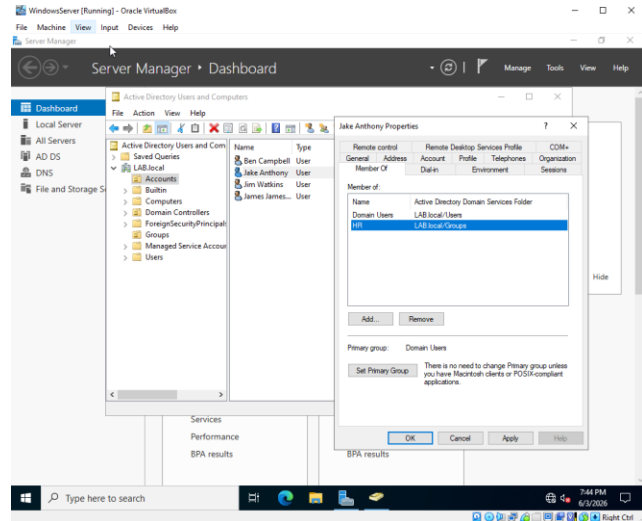
Permissions Are Audited

CRITICAL: Never give a user access to the wrong folder. Permission changes are logged and audited with timestamps. If someone gains unauthorized access to data because of an incorrect permissions assignment, it will be traced back to whoever made the change.

Part 1 – Add the User to the Correct Group

Before working with folders, the user needs to be in the correct Active Directory group.

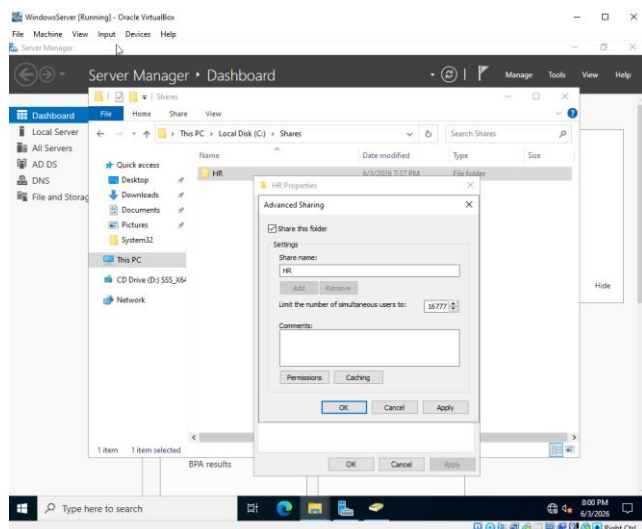
1. Open **Active Directory Users and Computers** and go to the **Accounts** OU.
2. Double-click **Jake Anthony** and go to the **Member Of** tab. Click **Add**.
3. Type **HR** and click **Check Names**. It will underline the name if the group is found.
4. Click **OK**, then **Apply**, then **OK**.
5. To verify, go to the **Groups** OU, double-click **HR**, and check the **Members** tab. Jake should appear in the list.



Best Practice: Always add users to groups rather than assigning permissions directly to individual users. When someone joins or leaves, you just update the group – never the folder permissions.

Part 2 – Create the Shared Folder Structure

6. On the Windows Server, open **File Explorer** and go to the **C: drive**.
7. Create a new folder and name it **Shares**. This will be the root folder for all department shared folders.
8. Inside **Shares**, create a folder named **HR**.
9. Inside **HR**, create subfolders for department data – for example:
 - **Resumes**
 - **Forms**
 - **Medical Records**



Note: To find the network path of a shared folder later, right-click the folder, go to Properties > Sharing tab. The network path will be displayed there after the folder has been shared.

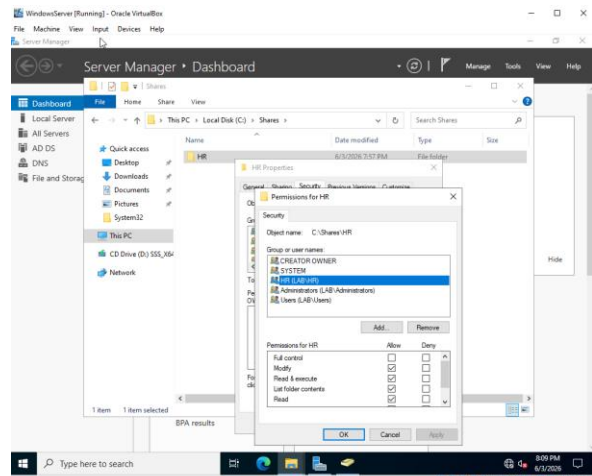
Part 3 – Configure Permissions

There are two layers of permissions to configure – Share Permissions and NTFS Permissions. Share permissions control access over the network. NTFS permissions control access at the file system level. Both need to be set.

Shared Permissions

10. Right-click the **HR** folder > **Properties** > **Sharing** tab > **Advanced Sharing**.
11. Check **Share this folder**, then click **Permissions**.
12. Under the Permissions for Everyone, check **Allow** for **Full Control, Change, and Read**. Click **Apply**, then **OK**.
13. Click **OK** to close Advanced Sharing and stay on Properties. Go to the **Security** tab.

Note: Setting Everyone to Full Control at the Share level is standard practice. The real access restriction happens at the NTFS level below.



NTFS Permissions

14. On the **Security** tab, click **Edit**, then **Add**.
15. Type **HR** and click **Check Names**. Click **OK**.
16. With HR selected, check **Allow** for **Modify**. Click **Apply**, then **OK**.

Note: Modify permission means HR group members can read, create, edit, and delete files inside the HR folder. Do not give Full Control to regular users – Full Control includes the ability to change permissions, which should only be for administrators.

Best Practice: Always assign NTFS permissions to Groups, never individual users.

Part 4 – Verify Access on the Client Machine

17. On the Windows 11 VM, log in as **LAB\JAnthony**.
18. If Jake was already logged in when permissions were changed, **log out and log back in** for the new group membership to take effect.
19. Open **File Explorer** and navigate to the HR shared folder to confirm access works.

Important: new group membership doesn't take effect until the user logs out and back in. This is one of the most common helpdesk calls after granting folder access.

Part 5 – Manual Network Driver Mapping

When a user asks, “I have access now, but where is the folder?”, the shared folder can be mapped as a drive letter on their machine for quick access.

20. On the Windows 11 VM logged in as Jake Anthony, open **File Explorer**.
21. Right-click **Network** in the left panel and select **Map network drive**.
22. Set the drive letter to **H:** for HR.
23. In the **Folder** field, enter the network path:

[\\LAB.LOCAL\HR](#)

24. Click **Finish**. The HR folder now appears as the H: drive in File Explorer.
25. Test it by creating a folder inside – it should reflect on the server in real time.

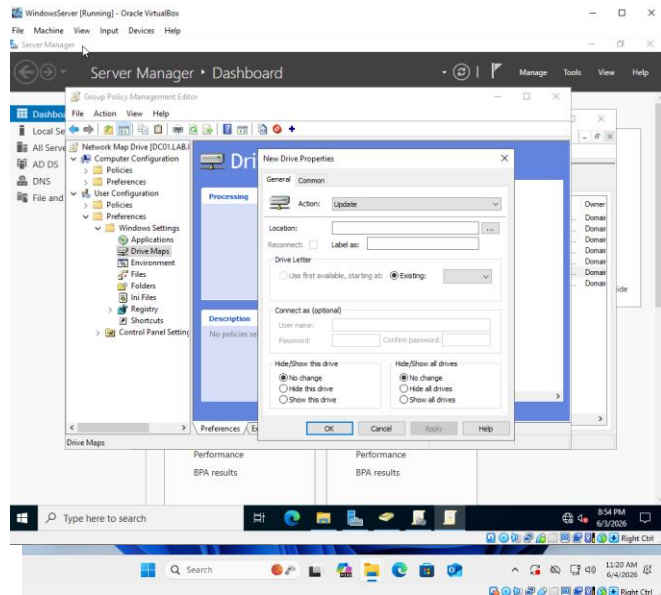
Best Practice: Using a matching drive letter convention is common in real environments – H for HR, F for Finance, I for IT. Makes it easy for users to remember.

Part 6 – Automating Network Drive Mapping via GPO

Manual drive mapping works, but it is not scalable in a real environment. If a company has hundreds of users, nobody should be mapping drives by hand. Group Policy can automate this so the correct drives appear automatically when a user logs in.

Create the GPO

26. On the Windows Server, go to **Tools > Group Policy Management**.
27. Expand **Forest: LAB.local > Domains > LAB.local**. Right-click **Group Policy Objects** and select **New**.
28. Name the policy **Network Map Drive** and click OK.
29. Right-click the new policy and select **Edit** to open the Group Policy Management Editor.



Configure the Drive Map

30. In the editor, go to:

User Configuration > Preferences >
Windows Settings > Drive Maps

31. Right-click **Drive Maps** and select
New > Mapped Drive.

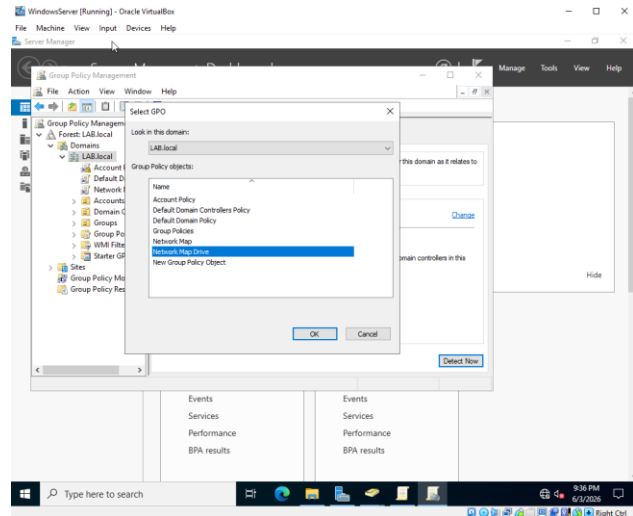
32. Set **Action** to **Create**.

33. In the **Location** field, enter the
network path of the shared folder:

\\DC01\HR

34. Set the drive letter to **H:** for HR.

35. Click **Apply**, then **OK**.



Set Up Item-Level Targeting

Without targeting, the mapped drive will show up for every user on the domain, regardless of which group they belong to. Item-Level Targeting fixes this by making the drive only appear for users in the correct group.

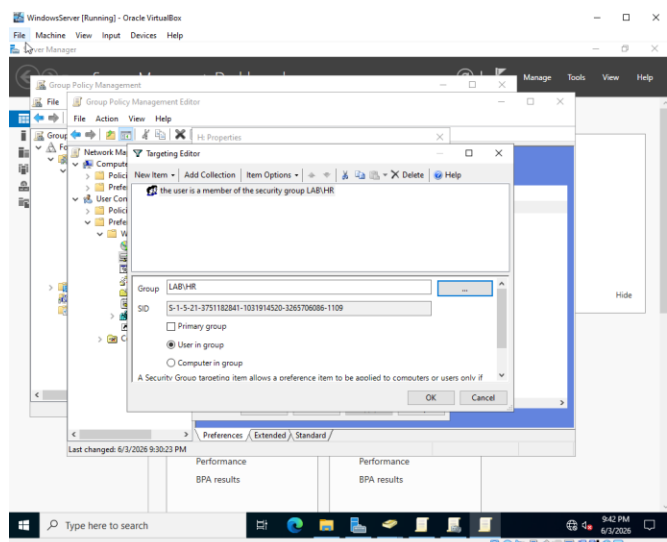
36. Double-click the drive map you just created.

37. Go to the **Common** tab and
check **Item-Level targeting**.
Click **Targetting**.

38. Click **New Item** and select
Security Group.

39. Click the "..." button next to the
Group field. Type **HR** and click
Check Names, then **OK**.

40. Click **OK** to close targeting, then
Apply and **OK**.



Note: Item-Level Targeting is what
makes the automation smart. Without it,
HR drives show up for IT users and vice
versa. Targeting ensures each other
only sees the drives they are supposed to have access to.

Link the GPO to the Domain

41. Back in Group Policy Management, right-click directly on **LAB.local** and select
Link an Existing GPO.

42. Select **Network Map Drive** and click **OK**.

43. The policy is now linked and will apply to all users in the domain, but the targeting
will ensure each drive only maps for the right group.

Testing the GPO

44. On the Windows 11 VM, **log out and log back in** as Jake Anthony.
45. Open **File Explorer** – the H: drive should appear automatically without any manual mapping.

Important: Use log out and log back in for testing GPO drive maps instead of gpupdate /force. Drive map preferences under User Configuration sometimes do not refresh correctly with gpupdate alone – a full log out and back in is more reliable.

Part 7 – Removing User Access

46. Open **Active Directory Users and Computers**.
47. Go to the **Groups** OU and double-click **HR**.
48. Click the **Members** tab, highlight the user, and click **Remove**.
49. Click **Apply**, then **OK**. Access is revoked immediately.

Best Practice: Because permissions are group-based, removing access is just removing the user from the group. No need to touch the folder or the GPO at all.

Key Takeaways

- **Shared folders belong on a File Server**, not the Domain Controller – keep them separate to avoid one failure taking down everything
- **Two permission layers** – Share Permissions open the network door, NTFS Permissions control what users can actually do inside the folder
- **Always assign permissions to Groups**, never individual users – makes managing access much easier when people join or leave
- **Modify vs Full Control** – regular users get Modify, Full Control is for administrators only
- **Log out and back in** – new group memberships and GPO drive maps need a fresh login to take effect
- **Item-Level Targeting** – essential for GPO drive maps. Without it, every user sees every drive regardless of their group
- **Permissions are audited** – every access change is logged with a timestamp. Giving the wrong person access to the wrong folder will be traced back to whoever made the change
- **GPO automation is the real-world standard** – no IT department manually maps drives for hundreds of users. GPO handles it automatically on every login

